

# Required practical: I–V characteristics

## Answers

Year 10 GCSE Physics • AQA Trilogy RP 16 • Physics RP 4

Name \_\_\_\_\_

Class \_\_\_\_\_ Date \_\_\_\_\_

Plan. Measure. Explain.

Work in pairs. Read the whole method before you build anything.

Online version with hints: [panphy.app/gcsephy/year10phy/unit01/practical/iv-characteristics/](https://panphy.app/gcsephy/year10phy/unit01/practical/iv-characteristics/)

## 1 Aim

Use circuit diagrams to construct circuits that investigate the I–V characteristics of a filament lamp, a diode and a resistor at constant temperature.

$$R = V \div I$$

R in ohms ( $\Omega$ ) · V in volts (V) · I in amperes (A)

## 2 Apparatus

- Variable d.c. power supply (0–6 V) or battery pack
- Variable resistor (rheostat)
- Ammeter (0–1 A) and milliammeter (0–100 mA), or multimeters
- Voltmeter (0–10 V)
- Switch and connecting leads
- Fixed resistor, for example 10  $\Omega$
- Filament lamp, for example 6 V 0.3 A
- Diode and a protective resistor, for example 100  $\Omega$

## 3 Safety

- The lamp gets hot. Do not touch it while it is on or just after.
- Do not go above the lamp's rated p.d. (for example 6 V).
- Always use the protective resistor with the diode, so the current stays small.

# Before you start

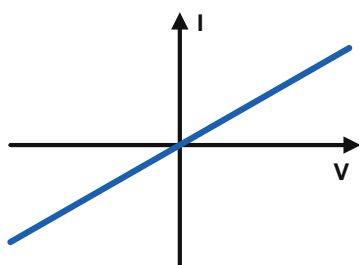
Learn the key words, then make a prediction before you build anything.

## 4 Key words

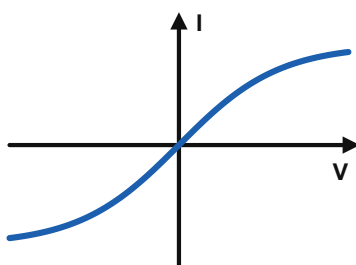
|                             |                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------|
| <b>Independent variable</b> | The variable you change.                                                                |
| <b>Dependent variable</b>   | The variable you measure.                                                               |
| <b>Control variable</b>     | A variable you keep the same, so the test is fair.                                      |
| <b>Resolution</b>           | The smallest change a meter can show, for example 0.01 A.                               |
| <b>Anomalous result</b>     | A result that does not fit the pattern of the others.                                   |
| <b>I–V characteristic</b>   | A graph of current (y-axis) against p.d. (x-axis) for a component.                      |
| <b>Ohmic conductor</b>      | A component whose current is directly proportional to the p.d. at constant temperature. |

## 5 Predict

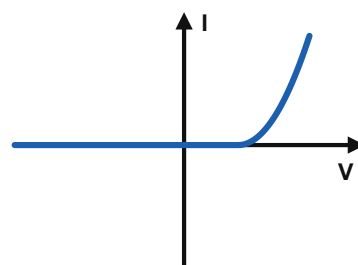
Sketch the I–V graph you expect for each component. Show both positive and negative p.d.



Fixed resistor



Filament lamp



Diode

Expected: a straight line through the origin; an S-shaped curve; no current in reverse, then a steep rise.

# Method Set up and measure

Method • I–V characteristics • Use the variable resistor to change the p.d. across each component. Record the current each time, then reverse the supply and repeat.

## 1 Variables

| Variable    | What it is                        | How                                         |
|-------------|-----------------------------------|---------------------------------------------|
| Independent | p.d. across the component         | Adjust the variable resistor or the supply  |
| Dependent   | Current through the component     | Read the ammeter                            |
| Control     | Temperature of the fixed resistor | Small currents; switch off between readings |

## 2 Method

1. Build the first circuit below with the fixed resistor as the test component. The voltmeter goes across the component only.

☐
2. Set the variable resistor to its highest resistance. Close the switch and record the p.d. and the current.

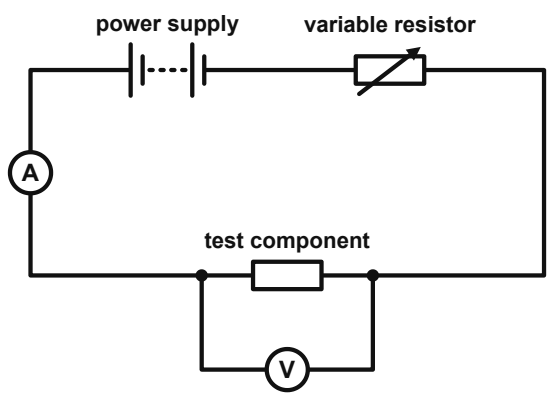
☐
3. Change the variable resistor to increase the p.d. in steps of about 1 V. Record the p.d. and current each time. Switch off between readings so the resistor stays cool.

☐
4. Swap the connections to the power supply so the current flows the other way. Repeat steps 2 and 3. Record these p.d. and current values as negative numbers.

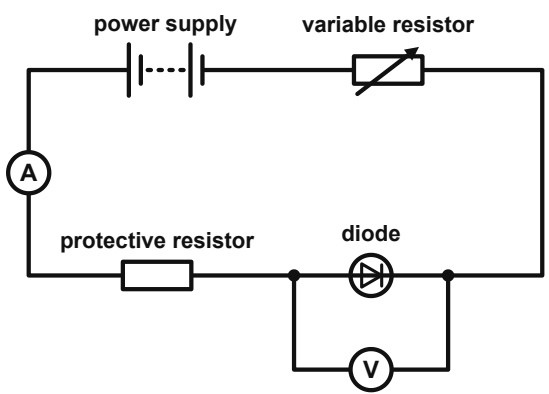
☐
5. Replace the resistor with the filament lamp and repeat steps 2 to 4. Do not go above the lamp's rated p.d.

☐
6. Build the second circuit for the diode. Use a milliammeter and increase the p.d. in small steps of about 0.1 V from 0.4 V, where the current starts to rise quickly. Then reverse the supply and take readings.

☐



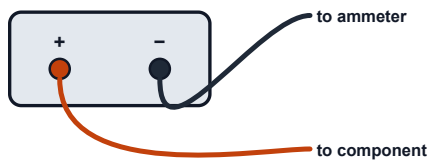
Test the resistor first, then swap in the filament lamp.



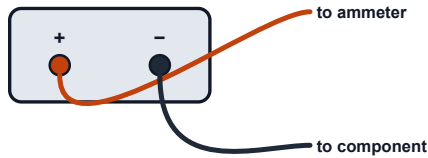
For the diode: add a protective resistor and use a milliammeter. The voltmeter goes across the diode only.

# Method Set up and measure

Forward: record + values



Reversed: swap the two leads



Swap the two leads at the power supply. The meters now show negative readings: write the minus sign.

real diode



symbol



It conducts when the current flows towards the band (the bar).

The band on the diode matches the bar in its symbol.  
Forward: current flows towards the band.

**Why reverse the supply?** An I–V graph shows both directions. Reversing the connections gives the negative half of the graph.

**Using the variable resistor.** Start at its highest resistance, so the p.d. across the component is small. Lower its resistance a step at a time to increase the p.d.

# Results Record your readings

Results • I–V characteristics • Record every reading as you take it. Reversed readings are negative: write the minus sign.

## 1 Results: fixed resistor

| Forward  |             | Reversed (negative) |             |
|----------|-------------|---------------------|-------------|
| p.d. / V | Current / A | p.d. / V            | Current / A |
| 1.0      | 0.10        | –1.0                | –0.10       |
| 2.0      | 0.20        | –2.0                | –0.20       |
| 3.1      | 0.31        | –3.0                | –0.30       |
| 3.9      | 0.39        | –4.0                | –0.40       |
| 5.0      | 0.50        | –5.1                | –0.51       |
| 6.0      | 0.60        | –6.0                | –0.60       |

## 2 Results: filament lamp

Calculate the resistance,  $R = V \div I$ , for each forward reading.

| Forward  |             |                       | Reversed (negative) |             |
|----------|-------------|-----------------------|---------------------|-------------|
| p.d. / V | Current / A | Resistance / $\Omega$ | p.d. / V            | Current / A |
| 0.5      | 0.08        | 6.3                   | –0.5                | –0.08       |
| 1.0      | 0.11        | 9.1                   | –1.0                | –0.11       |
| 2.0      | 0.16        | 12.5                  | –2.0                | –0.17       |
| 3.0      | 0.21        | 14.3                  | –3.0                | –0.20       |
| 4.5      | 0.26        | 17.3                  | –4.5                | –0.26       |
| 6.0      | 0.30        | 20.0                  | –6.0                | –0.30       |

# Results Record your readings

## 3 Results: diode

Use a milliammeter. Take small steps near 0.6 V.

| Forward  |              | Reversed (negative) |              |
|----------|--------------|---------------------|--------------|
| p.d. / V | Current / mA | p.d. / V            | Current / mA |
| 0.20     | 0.0          | -0.2                | 0.0          |
| 0.40     | 0.1          | -0.4                | 0.0          |
| 0.50     | 0.5          | -0.6                | 0.0          |
| 0.55     | 1.4          | -0.8                | 0.0          |
| 0.60     | 4.3          |                     |              |
| 0.65     | 13           |                     |              |
| 0.68     | 25           |                     |              |
| 0.70     | 40           |                     |              |

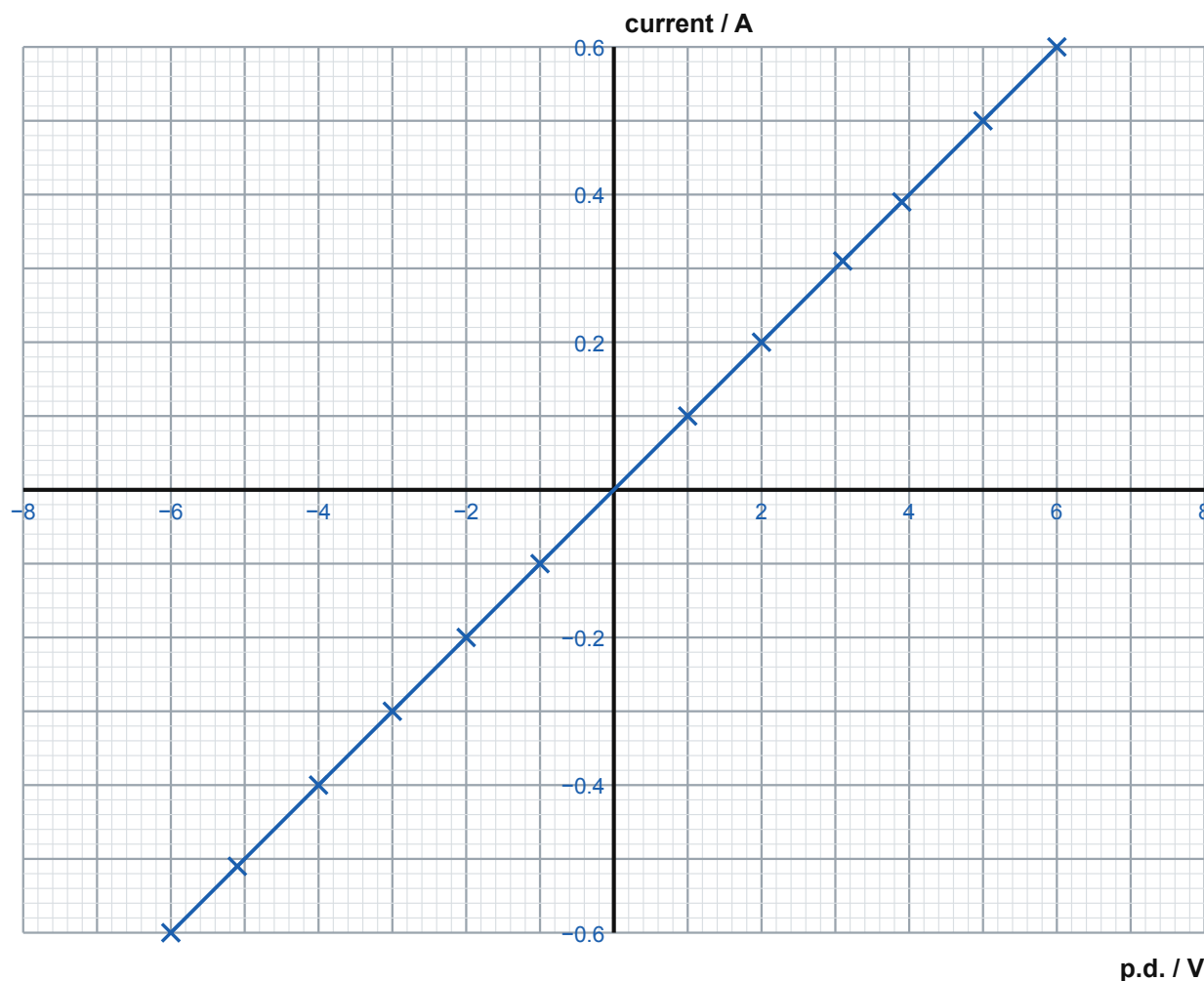
**Check your readings.** Every graph should pass through the origin: with no p.d. across a component, there is no current through it.

# Analysis Graphs and conclusions

Analysis • I–V characteristics • Plot current (y-axis) against p.d. (x-axis) for each component, using the positive and negative readings. Then explain each shape.

## 1 Fixed resistor graph

Plot current against p.d. for the fixed resistor. Add scales to both axes. Draw a line of best fit.



**Q1** Describe the I–V graph for the fixed resistor. What does it tell you about its resistance?

A straight line through the origin, the same shape in both directions. The current is directly proportional to the p.d., so the resistance stays constant. The resistor is an ohmic conductor at constant temperature.

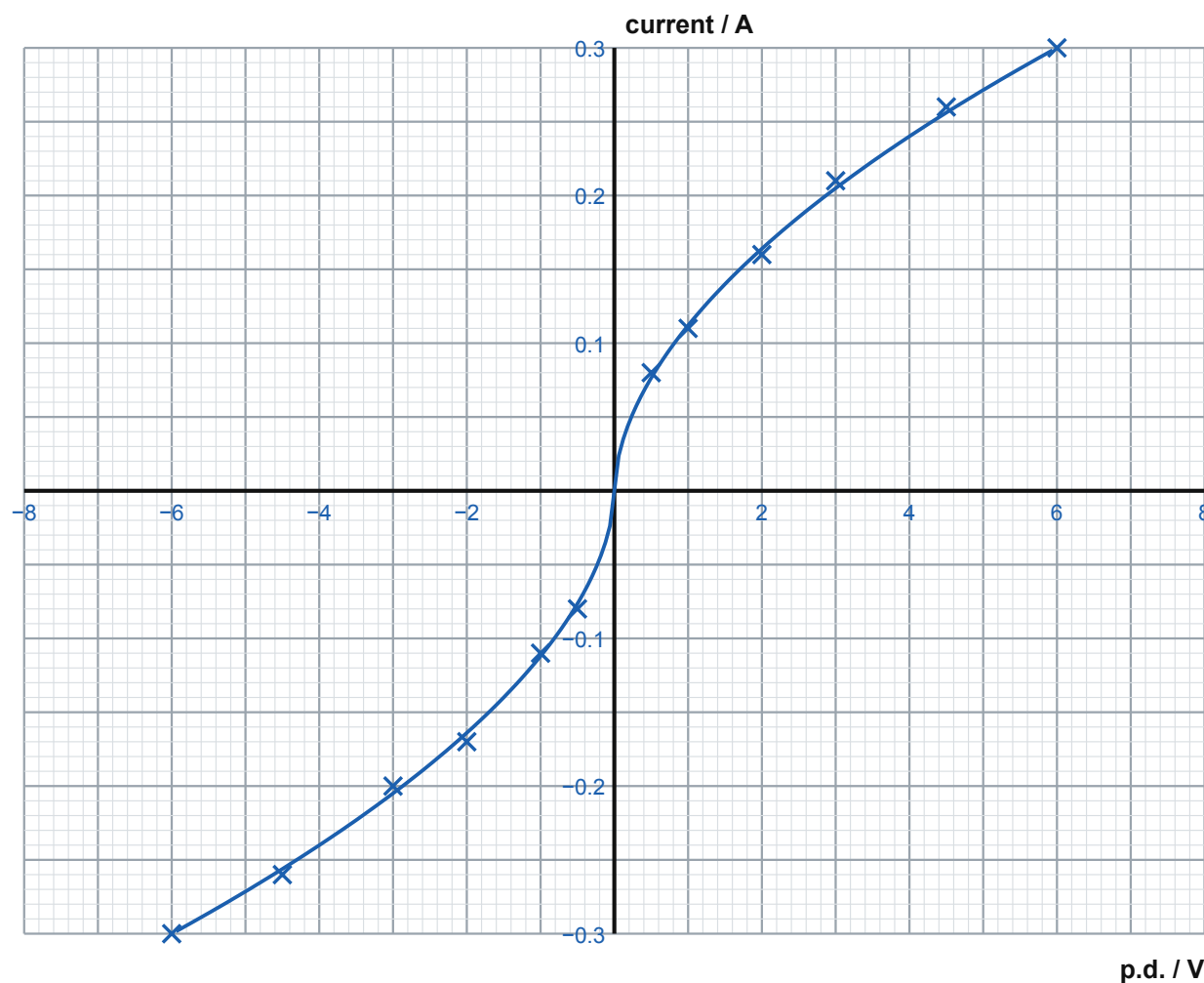
**Q2** Calculate the resistance of the fixed resistor using one pair of your readings.

For example,  $R = V \div I = 6.0 \div 0.60 = 10 \, \Omega$ .

# Analysis Graphs and conclusions

## 2 Filament lamp graph

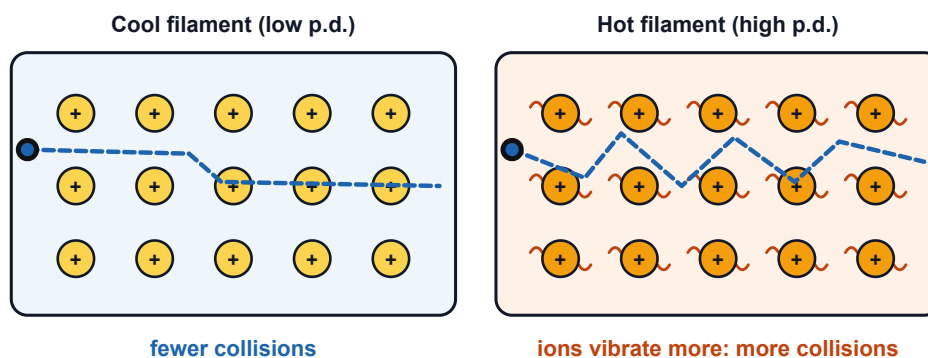
Plot current against p.d. for the filament lamp. Add scales to both axes. Draw a smooth curve of best fit.



**Q3** Use your readings to calculate the resistance of the lamp at your lowest and highest p.d. What happens to the resistance?

Example: at 0.5 V,  $R = 0.5 \div 0.08 \approx 6 \Omega$ . At 6.0 V,  $R = 6.0 \div 0.30 = 20 \Omega$ . The resistance of the lamp increases as the p.d. (and current) increases.

# Analysis Graphs and conclusions



Electrons (blue path) collide with the metal ions (+). A hotter filament has vibrating ions, so there are more collisions and a higher resistance.

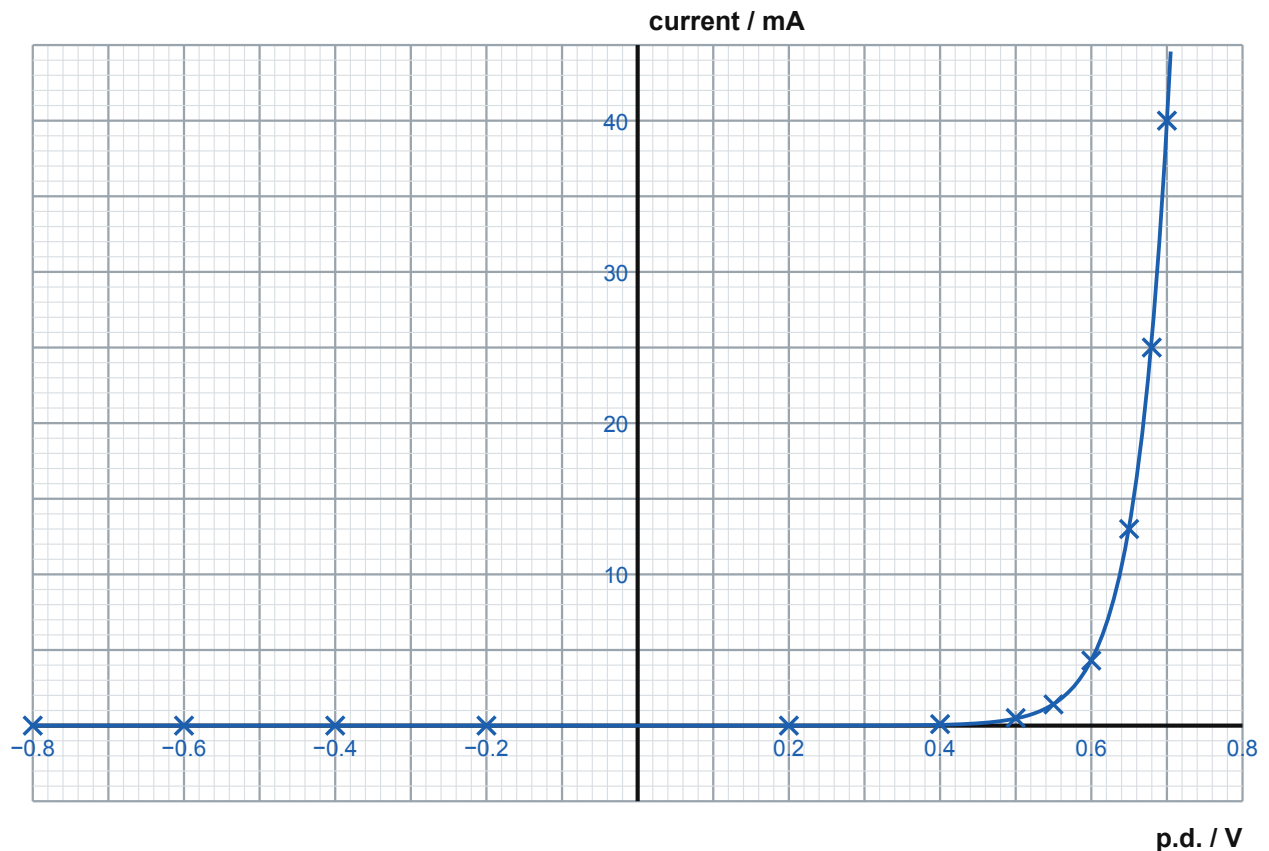
**Q4** Explain why the resistance of the filament lamp changes. **[3 marks]**

A bigger current heats the filament to a higher temperature (1). The metal ions vibrate more (1). Electrons collide with the ions more often, so the resistance increases (1).

# Analysis Graphs and conclusions

## 3 Diode graph

Plot current against p.d. for the diode. Add scales to both axes. Draw a smooth curve of best fit.



**Q5** Describe the I–V graph for the diode in the forward and reverse directions.

Reverse: the current is (almost) zero, because the diode has a very high resistance in the reverse direction.

Forward: there is almost no current until about 0.6 V; then the current rises very steeply.

**Q6** Explain why a protective resistor is connected in series with the diode.

Once the diode conducts, its resistance becomes very low. Without the protective resistor the current would become very large and could damage the diode, the meter or the supply. The resistor limits the current.

**Q7** Suggest why smaller steps in p.d. were used for the diode than for the other components.

The current changes very quickly over a small range of p.d. near 0.6 V. Small steps give more points there, so the shape of the curve can be drawn accurately.

# Analysis Graphs and conclusions

## 4 Compare the three components

Complete the table using your three graphs.

| Component      | Shape of the I–V graph                                | As the p.d. increases, the resistance...                | Ohmic? |
|----------------|-------------------------------------------------------|---------------------------------------------------------|--------|
| Fixed resistor | Straight line through the origin                      | stays the same                                          | Yes    |
| Filament lamp  | S-shaped curve through the origin                     | increases (the filament gets hotter)                    | No     |
| Diode          | Zero current in reverse; steep rise above about 0.6 V | is very high in reverse; falls quickly once it conducts | No     |

**Q8** A lamp has a p.d. of 4.5 V across it and a current of 0.26 A through it. Calculate its resistance.

$$R = V \div I = 4.5 \div 0.26 = 17 \, \Omega \, (17.3 \, \Omega).$$

**Q9** Why were readings also taken with the supply reversed?

To see whether the component behaves the same way when the current flows in the opposite direction. The resistor and lamp do; the diode does not, because it only lets current flow one way.

**Q10** At the highest p.d. the fixed resistor felt warm. Suggest how this would affect the top end of its I–V graph.

As the resistor warms up, its resistance increases, so the current is a little lower than expected. The top of the line bends slightly towards the p.d. axis instead of staying straight. Switching off between readings avoids this.

# Exam practice Plan the practical

Exam practice • I–V characteristics • A six-mark method question like the ones AQA asks about this practical. Write your answer without looking at the method.

**AQA-STYLE** **Q11** Describe a method to find how the current through a filament lamp depends on the p.d. across it. Include both positive and negative values of p.d. You may include a circuit diagram in your answer.

**[6 marks]**

## Marking points (level of response):

- Connect the lamp in series with an ammeter, a variable resistor and a power supply.
- Connect a voltmeter in parallel across the lamp.
- Adjust the variable resistor (or supply) to change the p.d.
- Record the p.d. and current for a range of values, for example in 1 V steps.
- Do not exceed the lamp's rated p.d.
- Reverse the connections to the power supply and repeat, recording negative values.
- Plot a graph of current (y-axis) against p.d. (x-axis) and draw a smooth curve.

Level 3 (5–6 marks): a clear, logical method that would give valid results. Level 2 (3–4 marks): most steps, but not fully logical or detailed. Level 1 (1–2 marks): some relevant steps.

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## 1 Check your answer

When you have finished, tick each point your method includes.

- |                                                                                   |                                                                                     |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <input type="checkbox"/> A circuit: lamp, ammeter and variable resistor in series | <input type="checkbox"/> Record the current at each p.d.; stay below the rated p.d. |
| <input type="checkbox"/> Voltmeter connected across the lamp only                 | <input type="checkbox"/> Reverse the supply and record negative values              |
| <input type="checkbox"/> Change the p.d. with the variable resistor, in steps     | <input type="checkbox"/> Plot current (y) against p.d. (x); draw a smooth curve     |